

A Graph-based Framework for Coverage Analysis in Autonomous Driving

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Motivation and Existing Approaches

Motivation

Coverage analysis is crucial to ensure safety and reliability of autonomous driving systems

Coverage argument: have the observed scenes exhausted the space of traffic scenes that can occur?

Existing arguments are typically collected **per coverage factor**, or up to 2–3 factor interactions

This work: take the **complete traffic scene** as the unit of coverage

- each scene → one graph

- nodes carry actor parameters, edges carry relation parameters

- arbitrarily complex scenes (varying numbers of actors, higher-order interactions)
captured in one object, not decomposed into isolated factors

Scope: completeness of observed scenes, *not* safety of the system

Existing Approaches

Scenario-based testing is the dominant paradigm (PEGASUS: functional / logical / concrete scenario levels)

Approval trap: statistically demonstrating superior safety needs billions of test km

→ motivates simulation and systematic coverage methods

Standardized coverage criteria: ISO 21448, UL 4600

Domain-specific coverage buckets (Foretellix)

Recent trend: represent traffic scenes as **graphs** for scalable, label-free analysis

- semantic scene-graph embeddings,
contrastive learning, heterogeneous scenario graphs

Gap: these graph representations are not designed specifically for coverage analysis → this work

Argoverse 2.0: large-scale real-world dataset

250,000 scenarios, 11 s each

tracked trajectories (vehicles, pedestrians, cyclists) + semantic map

subset of 17,000 training scenes used

CARLA 0.9.15: open-source simulator

six maps, 20–60 vehicles per run, mixed traffic

2,050 scenes of 11 s each

Traffic Scene Graph and Construction Algorithm

Defining a Traffic Scene Graph

Map graph G_{map} : nodes = lane segments
(geometry + semantics)

Map graph edges: **following**, **neighbor**, **opposite**

Actor graph G_{actor} : nodes = actors at one
timestep (lane, position, speed, type, lane-change
indicator)

Actor graph edges: relation type + path length

Relation types (semantic hierarchy):

Following/Leading, Neighbor, Opposite

Lane-change indicator ($\Delta t = 1$ s) captures
cut-in/cut-out within a single-timestep graph

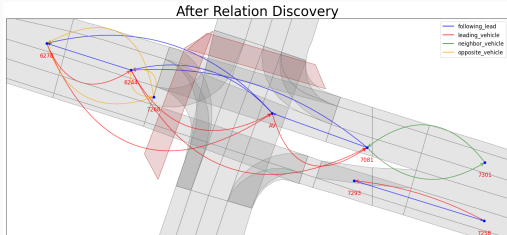
Hierarchical Graph Construction Algorithm

Phase 1 – Relation Discovery: find a connecting
path per actor pair; path structure sets relation
type; filtered by distance/path-length thresholds

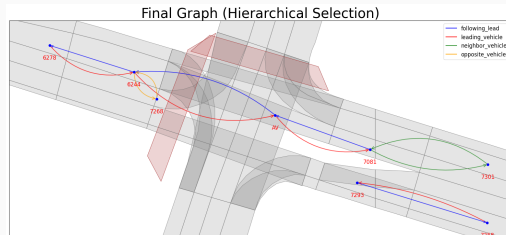
Phase 2 – Hierarchical Construction: add edges
leading/following $\rightarrow$ neighbor $\rightarrow$ opposite
(shortest path first); skip an edge if a path
already exists (BFS)

Relation	Dist. fwd/bwd [m]	Max. path [edges]
Leading/following	100	3
Neighbor	50 / 50	2
Opposite	100 / 10	2

Hierarchical Construction: Example



After relation discovery



After hierarchical selection

The discovery graph (left) contains all relations within distance limits. Hierarchical selection (right) removes redundant edges representable through existing paths.

Coverage Method 1: Subgraph Isomorphism

Traffic scene archetypes (lead-follow, cut-in, cut-out, opposite traffic, platoons, ...)
expressed as small graphs

Check whether a subgraph of scene graph G is isomorphic to archetype A (VF2 algorithm)

Strategy:

- define archetype set S

- fix node/edge attributes considered

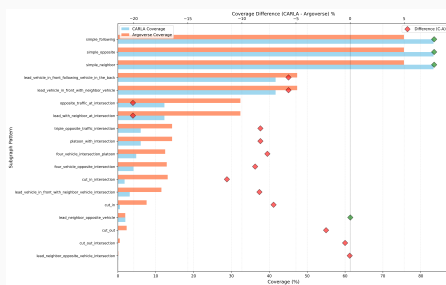
- match each scene graph against $S \rightarrow$ coverage table C

18 archetypes defined: 2-actor (3), 3-actor (9), 4-actor (4), 5-actor (2)

Archetypes are user-defined, illustrative—not a complete or safety-certified taxonomy

Real safety case: archetypes derived top-down (HARA / UL 4600), completeness of the set is critical

Results: Scenario Archetype Coverage Gaps



CARLA higher coverage only for the 3 simple 2-actor scenarios

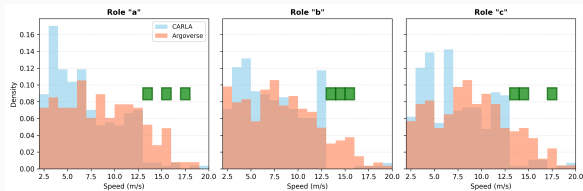
All multi-actor scenarios: Argoverse shows higher occurrence

cut_out_intersection: <1% CARLA vs. ~8% Argoverse

lead_neighbor_opposite_vehicle_intersection: ~6% Argoverse, virtually absent in CARLA

cut_in/cut_out: underrepresented in CARLA regardless of intersections

Results: Parameter Distribution Gaps



Role-specific speed distributions,
lead_vehicle_in_front_with_
neighbor_vehicle scenario

CARLA concentrates around 3–5 m/s
Argoverse extends to 15–20 m/s and
beyond

Coverage gap in the 13–17 m/s range
for all three roles

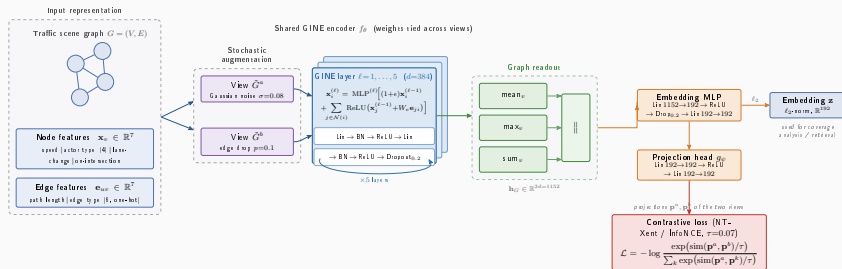
Gaps arise at the parameter level even
within structurally matching scenarios

Coverage Method 2: Graph Embeddings (GINE)

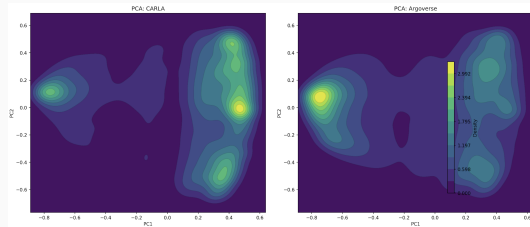
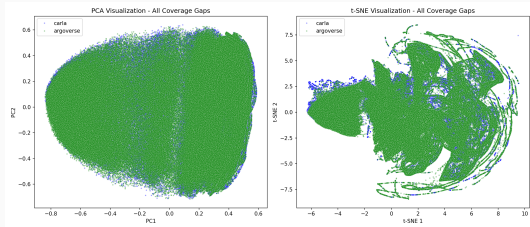
Real traffic scenes are far more complex than predefined archetypes → complementary top-down approach

Graph Isomorphism Network with Edge features (GINE): natively encodes edge attributes (relation type + distance)

Self-supervised contrastive training (NT-Xent/InfoNCE, $\tau = 0.07$) on CARLA + Argoverse jointly



Results: Embedding Space Analysis



Plausibility check: nearest neighbours in embedding space share actor configuration and road geometry

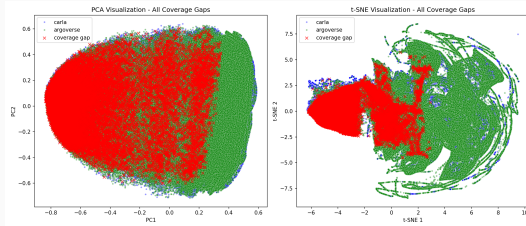
First two principal components explain 40% of the total variance

CARLA peaks at higher PC1 values

Argoverse peaks at negative PC1 values

→ datasets occupy different regions of the embedding space

Results: Coverage Gaps in Embedding Space



Argoverse graphs with under-represented archetype labels projected into embedding space and compared against CARLA density

23,148 Argoverse graphs identified as coverage gaps

Dominated by `cut_in` (~94% of Argoverse cut-in instances), `cut_out`, `cut_out_intersection`, `lead_neighbor_opposite_vehicle_intersection`

Gap graphs cluster where CARLA density is negligible → agreement between embedding and exact isomorphism labels

Used as a *coverage-screening* layer, not a standalone safety oracle

Summary and Future Work

Graph-based framework: hierarchical scene graphs combining map topology with actor relationships

Two complementary coverage methods:

- subgraph isomorphism against archetypes → interpretable, pattern-based coverage

- GINE embeddings via contrastive learning → similarity-based coverage, clustering, anomaly detection

Validated on Argoverse 2.0 and CARLA: reveals structural, parametric, and co-occurrence coverage gaps in simulation

Scales across scenarios without scenario-specific handling

Naturally accommodates varying numbers of actors

Future work:

- multi-timestep temporal graphs

- automatically extracting new archetype candidates from dense, unmatched embedding regions

Code: https://github.com/tmuehlen80/graph_coverage